MERIT EDUCATION GROUP

Minnesota's Curriculum. Built for Minnesota.

MN Standard 9.3.5.3, 9.3.5.7, 9.3.5.8

INTERMEDIATE ALGEBRA · UNIT 1: EXPRESSIONS · LESSON 02 · TEACHER LESSON PLAN

Translating Verbal Phrases

Day 2 of 20 · Duration: 50 minutes · Theme: Lake Superior & Duluth

MINNESOTA STANDARDS (MDE 2022)

Standard: 9.3.5.3

Apply knowledge of number systems extending from whole numbers to integers, from integers to rational numbers, from rational numbers to real numbers and from real numbers to complex numbers to solve equations. (MP2, MP7)

Standard: 9.3.5.7

Use the structure of an expression, equation and/or formula to create an equivalent form that is more helpful given the situation. Rearrange formulas to highlight a quantity of interest, using the same reasoning in solving equations. (MP6)

Standard: 9.3.5.8

Use the structure of an expression to write it in multiple ways.

ESSENTIAL QUESTIONS

- How do keywords like "more than" and "less than" change an expression?
- Why is the order of terms important when translating a phrase?
- How can the same verbal phrase be written in multiple ways?

VOCABULARY

Verbal Phrase — A phrase written in words that describes a mathematical situation. **Algebraic Expression** — A math phrase using variables, numbers, and operation symbols. **Translate** — To change words into an algebraic expression using math symbols. **Sum / Difference / Product / Quotient** — The results of adding, subtracting, multiplying, and dividing.

LEARNING OBJECTIVES & SUCCESS CRITERIA

- I can translate verbal phrases into algebraic expressions using variables and keywords.
- I can show my work step by step and check my answers for reasonableness.
- I can explain my reasoning to a partner using math vocabulary.

ENGAGEMENT STRATEGY

Strategy: Round Robin

Practice Format: Merit Time Game — Expression Race

Expectation: Every student responds on a whiteboard or to a partner before any answer is shared.

BELL RINGER — WHITEBOARDS READY!

Prompt: Translate “twice a number minus 4” into an algebraic expression. *Students write their response on a whiteboard and hold it up. Teacher scans for misconceptions before moving on.*

KICKOFF — STUDENT INTRO

Welcome, scholars! Over the next few weeks we will explore how letters and numbers work together to describe the world around us — from State Fair tickets to Ojibwe canoe-building and Somali-owned shops in Cedar-Riverside. This unit, Intermediate Algebra: Variables and Expressions, is your first step into algebra. Bring your whiteboard, your curiosity, and your willingness to make mistakes — mistakes are where the learning lives. Let’s get started!

LESSON PACING (50 MINUTES)

Phase 1: Bell Ringer 5 min min

Translate “twice a number minus 4” into an algebraic expression.

Phase 2: Direct Instruction 10 min min

Translating Verbal Phrases — teacher-led notes (I Do); GEMDAS reference as needed.

Phase 3: Guided Practice 12 min min

Work together (We Do) — Merit Time Game — Expression Race. Solve together: Write an expression for 5 more than a number. -> Answer: n + 5.

Phase 4: Culture Connection 8 min min

Somali (Somalia) — Somali Americans are a large African immigrant community in MN (Minneapolis/Saint Paul); C (with math word problem)

Phase 5: Practice & Game 10 min min

GO: Write an expression for 5 more than a number. | SET: Twice a number decreased by 7. One-third of a number. | EXTRA: The sum of twice a number and 9.

Phase 6: Exit Ticket 5 min min

Translate: "the quotient of a number and 6, decreased by 2."

PRACTICE PROBLEMS (GO / SET / EXTRA)

GO: Write an expression for 5 more than a number. Answer: n + 5

SET: Twice a number decreased by 7. One-third of a number. Answer: 2n - 7 and n/3

EXTRA: The sum of twice a number and 9. Answer: 2n + 9

CULTURE CONNECTION — MINNESOTA STORIES

> **Spotlight:** Somali (Somalia)
>
> Somali Americans are a large African immigrant community in MN (Minneapolis/Saint Paul); Cedar-Riverside = 'Little Mogadishu'.
>
> **Math Word Problem:** Somali herding tradition: Ibrahim counts x goats in the east pasture and 7 more in the west. Write an expression for the total goats, then find it when x = 5.

CAREER SPOTLIGHT — MINNESOTA

Career: Maurices — Duluth | Merchandise Planner | BS | $65,653 *MN DEED 25th-percentile wage; OES-cited. Connection: algebra skills used daily on the job.*

DIFFERENTIATION

• Scaffolding: sentence starters for translation tasks; GEMDAS reference card on desks.

• Extension: students write and solve their own word problems using the lesson's operation keywords.

• Language support: bilingual vocabulary cards; pair students with a math-language peer.

• Grouping: flexible pairs/trios by readiness; Merit Time games rotate roles each round.

ASSESSMENT

• Formative: whiteboard checks during the bell ringer; circulation during GO/SET/EXTRA; exit ticket each lesson.

• Game-based: Merit Time games give instant fluency feedback (L1–L5).

• Summative: Quiz 1 (Day 7, L1–L5), Quiz 2 (Day 17, L6–L15), End-of-Unit Test (Day 20, L1–L17).